UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/233,319	04/16/2021	Yu Fang	SONA-0006-U01-C07	7216

87084 7590 07/07/2021

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
LIN, JSING FORNG	

ART UNIT	PAPER NUMBER
2446	

NOTIFICATION DATE	DELIVERY MODE
07/07/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/233,319	Applicant(s) Fang et al.	
	Examiner JSING FORNG LIN	Art Unit 2446	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 4/16/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-46 is/are pending in the application.

5a) Of the above claim(s) 28-34 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-27 and 35-46 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☑ Claim(s) 1-46 are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☑ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

This Action is in response to application filed on 4/16/2021.

Claims 28-34 are withdrawn.

Claims 1-27, 35-46 are pending in this application.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 4/16/2020 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Election/Restrictions

Applicant's election without traverse of group I, claims 1-27, 35-46, in the telephone communication on 6/21/2021 is acknowledged.

Double Patenting

A rejection based on double patenting of the "same invention" type finds its support in the language of 35 U.S.C. 101 which states that "whoever invents or discovers any new and useful process... may obtain a patent therefor..." (Emphasis added). Thus, the term "same invention," in this context, means an invention drawn to

identical subject matter. See Miller v. Eagle Mfg. Co., 151 U.S. 186 (1894); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Ockert, 245 F.2d 467, 114 USPQ 330 (CCPA 1957).

A statutory type (35 U.S.C. 101) double patenting rejection can be overcome by canceling or amending the claims that are directed to the same invention so they are no longer coextensive in scope. The filing of a terminal disclaimer cannot overcome a double patenting rejection based upon 35 U.S.C. 101.

Claim 1 are provisionally rejected under 35 U.S.C. 101 as claiming the same invention as that of claims 1 of copending Application No. 17,027,167. This is a provisional statutory double patenting rejection since the claims directed to the same invention have not in fact been patented.

U.S. Application No. 17,027,167	Present Application No. 17,233,319
Claim 1. A system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising:	Claim 1. A system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising:

a policy manager circuit structured to interpret a policy comprising a network regulation description;	a policy manager circuit structured to interpret a policy comprising a network regulation description;
a configuration circuit structured to configure at least one network interface circuit in response to the policy; and	a configuration circuit structured to configure at least one network interface circuit in response to the policy; and
the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.	the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., In re Berg, 140 F.3d 1428, 46 USPQ2d

1226 (Fed. Cir. 1998); In re Goodman, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); In re Longi, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); In re Van Ornum, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Thorington, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims 24 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1 of co-pending U.S. Application No. 17,233,324. Although the claims at issue are not identical, they are not patentably distinct from each other because claims in the co-pending application anticipated claims in the present application.

US Application No. 17,233,324	Present Application No. 17,233,319
Claim 1. A system comprising: a vehicle having an Ethernet based network and a controller area network (CAN) based network; a CAN vehicle control device disposed onboard the vehicle, connected to the CAN based network, and structured to control operation of a component of the vehicle; an Ethernet vehicle control device disposed onboard the vehicle, connected to the Ethernet based network, and structured to electrically communicate with the CAN vehicle control device; an Ethernet switch disposed onboard the vehicle and having a plurality of physical ports connected to the Ethernet based network, at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device; a CAN gateway disposed onboard the	Claim 24. A system comprising: a vehicle having a first network and a second network; a first device on the first network; and a converged network device (CND) interposed between the first network and the second network and structured to facilitate communications between the first device and the second network; wherein the first network is of a different type than the second network; wherein the first network is a Controller Area Network (CAN) based network and the second network is an Ethernet based network; wherein the CND comprises: a configurable edge gateway (CEG) structured to access the CAN based network; an Ethernet switch structured to access to the Ethernet based network; and wherein the CEG provides a message to the Ethernet switch in

vehicle and connected to the CAN based network and the Ethernet switch; and a network convergence circuit defined at least in part by the Ethernet switch and/or the CAN gateway, and structured to facilitate electronic communications between the Ethernet vehicle control device and the CAN vehicle control device by translating and relaying data messages between the Ethernet based network and the CAN based network.	response to a corresponding message on the CAN based network.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to

point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-5, 7-9, 35-46 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1) in view of Roese et al (US 2009/0187968 A1).

Regarding claim 1, SCHOCH teaches a system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone (SCHOCH fig. 1 vehicle internal network with plurality security zones communicating with different message type; [0023][0032]);

a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured a policy comprising a network regulation description (SCHOCH [0047][0049] a domain controller may provide a firewall of separating two security zones based on predefined security criterion [0023]);

a configuration circuit structured to configure at least one network interface circuit in response to the policy (SCHOCH [0023] a domain controller is configured to transmit message between two security zones if the messages meet criterion; claim 10); and

the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone (SCHOCH [0023] control

device on different security zone communicate each other if predefined criteria meet; [0028][0032] translate message between zones or e.g. a rate or frequency of a given message type may be required within a predetermined value interval).

However, SCHOCH does not explicitly teach a circuit structured to interpret a policy comprising a network regulation description (Note: SCHOCH teaches a domain controller with a firewall is used to separate zone based on security criterion as recite above);

Roese teaches a circuit structured to interpret a policy comprising a network regulation description and to configure at least one network interface circuit in response to the policy (Roese fig. 3 item #306 determine dynamic polices; item #309 set polices; [0034] determine policies and set polices of the attached connection);

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Roese to interpret a policy comprising a network regulation description and to configure at least one network interface circuit in response to the policy.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform operation more efficiently, i.e. update policy dynamically (Roese [0034]).

Regarding claim 2, SCHOCH-Roese teaches the system of claim 1, wherein the policy manager circuit is further structured to receive a policy communication from an external device, and to interpret the policy by performing, in response to the policy

communication, one of storing the policy or updating a stored policy (Roese [0031] acquire policy from remote policy server; [0034] saving policies).

Regarding claim 3, SCHOCH-Roese teaches the system of claim 2, wherein the external device is communicatively coupled to the policy manager circuit through at least one of the first network zone or the second network zone (SCHOCH [0039][0042] external server or smart phone couple to vehicle network domain controller).

Regarding claim 4, SCHOCH-Roese teaches the system of claim 2, wherein the external device is communicatively coupled to the policy manager circuit via at least one of a wireless network connection or a cellular network connection (SCHOCH [0039][0042] wireless connection).

Regarding claim 5, SCHOCH-Roese teaches the system of claim 2, wherein the policy manager circuit is further structured to verify the policy before performing the one of storing the policy or updating the stored policy (Roese [0033-0034] when the policy has been determined, save policy).

Regarding claim 7, SCHOCH-Roese teaches the system of claim 1, wherein the policy comprises at least one data collection parameter (Roese fig. 5 policy include port speed, bandwidth limit etc.; SCHOCH [0032]).

Regarding claim 8, SCHOCH-Roese teaches the system of claim 1, wherein the policy comprises a permission value corresponding to at least one end point of the first network zone or the second network zone, wherein the permission value comprises at least one permission value selected from the values consisting of: a data collection permission value; a service publication permission value; a service subscription permission value; or an external communication permission value (SCHOCH [0032] safety criterion include meet the predefined type spec).

Regarding claim 9, SCHOCH-Roese teaches the system of claim 1, wherein the policy is communicated to the CND from an external device (SCHOCH [0039], Roese [0031]).

Regarding claim 35, SCHOCH teaches a system, comprising: a vehicle having a first network zone and a second network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising (SCHOCH fig. 1, [0023][0032]): a policy manager circuit structured a policy comprising a network regulation description, wherein the policy comprises a default policy value (SCHOCH [0047][0049] a domain controller may provide a firewall of separating two security zones based on predefined security criterion [0023]); a configuration circuit structured to configure at least one network interface circuit in response to an external data routing description and an external data service description (SCHOCH [0023] a domain controller is configured to transmit message between two security zones if the messages meet criterion; claim 10; [0028] to route

and translate message between zones); the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone (SCHOCH [0023] control device on different security zone communicate each other if predefined criteria meet; [0028][0032]).

However, SCHOCH does not explicitly teach a circuit structured to interpret a policy comprising a network regulation description; (Note: SCHOCH teaches a domain controller with a firewall is used to separate zone based on security criterion as recite above)

Roese teaches a circuit structured to interpret a policy comprising a network regulation description and to configure at least one network interface circuit in response to the policy; (Roese fig. 3 item #306 determine dynamic polices; item #309 set polices; [0034] determine policies and set polices of the attached connection)

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Roese to interpret a policy comprising a network regulation description and to configure at least one network interface circuit in response to the policy.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform operation more efficiently, i.e. update policy dynamically (Roese [0034]).

Regarding claim 36, SCHOCH-Roses teaches the system of claim 35, wherein the policy further comprises a primary policy value, and wherein the configuration circuit

is further structured to configure the at least one network interface circuit in response to the primary policy value (Roeses [0038-0039] determine static policy and set policy [0034]).

Regarding claim 37, SCHOCH-Roese teaches the system of claim 36, wherein the configuration circuit is further structured to determine whether the primary policy value is present, and to utilize the primary policy value as the policy instead of the default policy value in response to determining the primary policy value is present (Roese [0012], claim 41, determine policy exist and apply policy).

Regarding claim 38, SCHOCH-Roese teaches the system of claim 36, wherein the policy further comprises a secondary policy value, and wherein the configuration circuit is further structured to configure at least one network interface circuit in response to the secondary policy value (Roese [0032-0033] determine static policy and dynamic policy and set policy).

Regarding claim 39, SCHOCH-Roese teaches the system of claim 38, wherein the configuration circuit is further structured to determine whether the secondary policy value is present, and to utilize the secondary policy value as the policy instead of either the default policy value or the primary policy value, in response to determining the secondary policy value is present (Roese [0012][0033] determine dynamic policy and set policy, based on policy exit, claim 41).

Regarding claim 40, SCHOCH-Roese teaches the system of claim 38, wherein the configuration circuit is further structured to apply the secondary policy value as the policy, and to apply consistent portions of the primary policy value as a further portion of the policy (Roese [0025][0036] policy may be restricted to portion of attached functions).

Regarding claim 41, SCHOCH-Roese teaches the system of claim 36, wherein the default policy value comprises a permanent storage policy, and wherein the primary policy value comprises a tool supplied policy (Roese [0044] policy includes storage limit, [0041] based on application associate with the attached function).

Regarding claim 42, SCHOCH-Roese teaches the system of claim 41, wherein the policy manager circuit is further structured to interpret the tool supplied policy from a second external device (Roese [0031] acquire policy from remote server and set policy, SCHOCH [0043] exchange message with external terminal).

Regarding claim 43, SCHOCH-Roese teaches the system of claim 42, wherein the second external device comprises at least one device selected from the devices consisting of: a service tool, a manufacturing tool, a web based application, and a cloud based application (Roeses [0031] external policy or application server, SCHOCH [0043]).

Regarding claim 44, SCHOCH-Roese teaches the system of claim 38, wherein the default policy value comprises a permanent storage policy, and wherein the primary

policy value comprises a tool supplied policy (Roese [0041][0044] based on size or storage limit).

Regarding claim 45, SCHOCH-Roese teaches the system of claim 44, wherein the secondary policy value comprises a post- manufacturing policy (Roese fig. 4 based on device type or i.e. 802.1X).

Regarding claim 46. The system of claim 45, wherein the policy manager circuit is further structured to interpret the tool supplied policy from a second external device (Roese [0031], SCHOCH [0043]).

Claims 6 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1) in view of Roese et al (US 2009/0187968 A1) as applied to claims 11-5, 7-9, 35-46 above and further in view of Penilla et al (US 2015/0210287 A1).

Regarding claim 6, SCHOCH-Roese teaches the system of claim 5, as recited above.

However, SCHOCH-Roese does not explicitly teaches wherein the policy manager circuit is further structured to provide a notification to the external device in response to the verifying the policy. (Note: Roese discloses [0023][0036] administrator monitor change triggers of attached connection and policy function 108 can be located at remote policy server)

Penilla teaches wherein the policy manager circuit is further structured to provide a notification to the external device in response to the verifying the policy. (Penilla [0140] notification is triggered to the user's mobile device if violation in the restrictions are detected)

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH-Roese in view of to wherein the policy manager circuit is further structured to provide a notification to the external device in response to the verifying the policy.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform operation more efficiently (Penilla [0140]).

Claims 10-23 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1) in view of Kim et al (US 2016/0255154 A1).

Regarding claim 10, SCHOCH teaches a system, comprising: a first network zone of a vehicle comprising a first interconnected plurality of end points; a second network zone of the vehicle comprising a second interconnected plurality of end points; (SCHOCH fig. 1 vehicle internal network with plurality security zones communicating with different message type; [0023][0032]); a converged network device (CND) interposed between the first network zone and the second network zone, wherein the CND is configured to regulate communications between end points of the first network zone and the second network zone; a first vehicle controller on the first network zone; a

second vehicle controller on the second network zone (SCHOCH [0047][0049] a domain controller may provide a firewall of separating two security zones based on predefined security criterion [0023][0028]); a network redundancy circuit structured to selectively provide a regulation control command; and wherein the CND is further configured to regulating the communications between end points of the first network zone and end points of the second network zone in response to the regulation control command (SCHOCH [0028][0032] message filter is to distinguish message and specify command to block or allow communication between zone based on safety criterion).

However, SCHOCH does not explicit teach wherein the CND is further configured to adjust the regulating the communications between zones (Note: SCHOCH disclose regulate message between zone as recite above).

Kim teaches wherein the CND is further configured to adjust the regulating the communications between zones (Kim [0091][0094] determine whether to update the security level and update the security level if needed).

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Kim to wherein the CND is further configured to adjust the regulating the communications between zones.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform data operation more efficiently (Kim [0091][0094]).

Regarding claim 11, SCHOCH-Kim teaches the system of claim 10, wherein the regulation control command comprises an off-nominal condition corresponding to the first vehicle controller (Kim [0077] risk assessment includes damage occur or security attack etc.).

Regarding claim 12, SCHOCH-Kim teaches the system of claim 11, wherein the off-nominal condition comprises a loss of a data element, and wherein the adjusted regulating the communications comprises providing an alternate data element to the first vehicle controller (Kim [0002][0102] failure occurs in an electronic part or security attack of a vehicle, [0091][0094] determine update and update security level)

Regarding claim 13, SCHOCH-Kim teaches the system of claim 11, wherein the off-nominal condition comprises a lost control function of the first vehicle controller, and wherein the adjusted regulating the communications comprises providing a data element corresponding to the control function of the first vehicle controller to the second vehicle controller (Kim [0091][0094] when the updating is determined to be needed, updating the security level, [0102] security level associate with parts failure or security attack).

Regarding claim 14, SCHOCH-Kim teaches the system of claim 13, wherein the lost control function comprises one of a partial loss or a full loss of the control function (Kim [0091][0094] update security level, [0102]).

Regarding claim 15, SCHOCH-Kim teaches the system of claim 13, wherein the lost control function utilizes a data value provided by an end point on the first network zone, and wherein the adjusted regulating the communications comprises providing the data value to the second vehicle controller (Kim [0035][0094] based on assessment value, SCHOCH [0028]).

Regarding claim16, SCHOCH-Kim teaches the system of claim 13, wherein the lost control function utilizes a data value provided by an end point on the second network zone, and wherein the adjusted regulating the communications comprises suppressing communication of the data value to the first network zone (Kim [0035][0094] update security level; SCHOCH [0028] filter message and specify control command between zones).

Regarding claim 17, SCHOCH-Kim teaches the system of claim 16, wherein: regulating the communications comprises processing a data value from an end point on the second network zone into a processed data value, and providing the processed data value to the first vehicle controller; wherein the lost control function utilizes the processed data value; and wherein the adjusted regulating the communications comprises providing the processed data value to the second vehicle controller (SCHOCH [0028][0032], Kim [0035][0091][0094]).

Regarding claim 18, SCHOCH-Kim teaches the system of claim 10, wherein the regulation control command comprises a loss of function of the first vehicle controller (Kim [0102], SCHOCH [0028]).

Regarding claim 19, SCHOCH-Kim teaches the system of claim 10, wherein the first vehicle controller is positioned in a first risk exposure profile, wherein the second vehicle controller is positioned in a second risk exposure profile, and wherein the first risk exposure profile is distinct from the second risk exposure profile (Kim [0077] risk assessment based on different security level).

Regarding claim 20, SCHOCH-Kim teaches the system of claim 10, wherein the first vehicle controller is positioned in a first risk exposure profile; the second vehicle controller is positioned in a second risk exposure profile; and the first exposure profile is distinct from the second risk exposure profile, wherein the distinction between the first risk exposure profile and the second risk exposure profile comprises a geometric distinction. (Kim [0077] include different risk level with different level security etc.)

Regarding claim 21, SCHOCH-Kim teaches the system of claim 10, wherein the first vehicle controller is positioned in a first risk exposure profile; the second vehicle controller is positioned in a second risk exposure profile; and the first exposure profile is distinct from the second risk exposure profile, wherein the distinction between the first risk exposure profile and the second risk exposure profile comprises an environmental distinction. (Kim [0077] different risk level includes vehicle part failure or security attack)

Regarding claim 22, SCHOCH-Kim teaches the system of claim 10, wherein the regulation control command comprises an off-nominal condition corresponding to the first network zone (SCHOCH [0028], Kim [0094]).

Regarding claim 23, SCHOCH-Kim teaches the system of claim 22, wherein the off-nominal condition comprises at least one condition selected from the conditions consisting of: a loss of communication between at least one end point of the first network zone and the first network zone; a physical failure of at least a portion of the first network zone; and a bandwidth limitation of the first network zone. (SCHOCH [0028][0032] rate limit, SCHOCH [0077] vehicle part failure)

Claims 24-27 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1) in view of Akella et al (US 2020/0036717 A1).

Regarding claim 24, SCHOCH teaches a system comprising: a vehicle having a first network and a second network; a first device on the first network; and a converged network device (CND) interposed between the first network and the second network and structured to facilitate communications between the first device and the second network; wherein the first network is of a different type than the second network (SCHOCH fig. 1 vehicle internal network with plurality security zones communicating with different message type; [0023][0032]); wherein the first network is a Controller Area Network (CAN) based network and the second network is an Ethernet based network (SCHOCH

[0031] domain controller can be Ethernet or CAN gateway / switch; [0028] exchange/translate message between different protocol zones); wherein the CND comprises: a configurable edge gateway (CEG) structured to access the CAN based network; an Ethernet structured to access to the Ethernet based network; and wherein the CEG provides a message to the Ethernet in response to a corresponding message on the CAN based network (SCHOCH [0040][0043] exchange message between different network zone, i.e. can be CAN and Ethernet, [0028] control unit translate message between zone).

However, SCHOCH does not explicitly teach an Ethernet switch structured to access to the Ethernet based network; and wherein the CEG provides a message to the Ethernet switch in response to a corresponding message on the CAN based network (Note: SCHOCH disclose domain controller can be Ethernet gateway or router / switch as recited above);

Akella teaches an Ethernet switch disposed onboard the vehicle at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device and CAN gateway (Akella fig. 5 [0050] Ethernet switch interconnect with IP gateway and CAN gateway).

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Akella to an Ethernet switch disposed onboard the vehicle at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device and CAN gateway.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform network interconnect more efficiently (Akella [0050]).

Regarding claim 25, SCHOCH-Akella teaches the system of claim 24, wherein the CEG is structured to provide the message to the Ethernet switch as an Ethernet message including an encapsulated CAN message based on the corresponding message on the CAN based network (Akella [0036-0037] encapsulate /decapsulate between CAN and Ethernet network).

Regarding claim 26, SCHOCH-Akella teaches the system of claim 24, wherein the CEG is positioned in a first housing, and wherein the Ethernet switch is positioned in a second housing (Akella [0050]).

Regarding claim 27, SCHOCH-Akella teaches the system of claim 24, wherein the CEG is disposed on a first board, and wherein the Ethernet switch is positioned on a second board (Akella [0050]).

Additional References

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

a. McKeefery et al., US 20200252339 A1: SYSTEM, METHOD AND APPARATUS FOR RESTRICTING USE OF A NETWORK DEVICE THROUGH AUTOMATED POLICY ENFORCEMENT
b. SCHMIDT et al., US 20180270230 A1: MOTOR VEHICLE COMMUNICATION NETWORK WITH SWITCH DEVICE
c. Joshi et al., US 20200267080 A1: SYSTEMS AND METHODS FOR DATA PROCESSING AND STORAGE IN VEHICLES HAVING A ZONE-BASED, CENTRAL COMPUTING IN-VEHICLE COMMUNICATIONS NETWORK ARCHITECTURE
d. Shalev et al., US 20190260800 A1: CRYPTIC VEHICLE SHIELD.
e. King et al., US 10650621 B1: Interfacing with a vehicular controller area network.
f. TORISAKI et al., US 2020/0145252 A1: In-Vehicle relay device, in-vehicle monitoring device, in-vehicle network system, communication monitoring method, and recording medium.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEVE LIN whose telephone number is 571-272-5137. The examiner can normally be reached on Monday – Friday 7:30 AM – 5:00 PM.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Brian Gillis can be reached on 571-272-7952. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/S. L./
Examiner, Art Unit 2446

/BRIAN J. GILLIS/
Supervisory Patent Examiner, Art Unit 2446